

Financial Econometrics

Event Study

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Did Elon Musk's tweet about taking Tesla private affect its stock price

On 7.8.2018, the CEO of Tesla, Elon Musk, tweeted: "Am considering taking Tesla private at \$420. Funding secured". On 6.8.2018 the stock was trading at \$341.99 and subsequently increased to \$370.34 on 08.08.2018, an increase of 8.3%.

It was unclear how serious Elon Musk's tweet was. Apart from that, it is illegal for companies and executives to give shareholders misleading information about potentially meaningful corporate events. Following the events, the Securities and Exchange Commission announced that they will sue Elon Musk. Their main area of concern was that Elon Musk's tweet triggered a trading frenzy which in turn led to a substantial increase in the stock price.

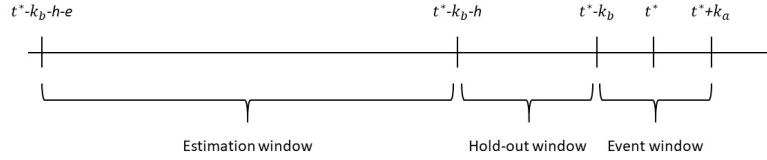
End of September, a settlement was reached between Musk, Tesla, and the Securities and Exchange Commission. Elon Musk stepped down as Tesla chairman (although he stayed on as CEO) and paid a fine of \$20 million, as did Tesla.

Data provided for this assignment

- Daily market and other factor returns, *factor_returns.txt*
 - *date*: date
 - *mkt.rf*: market return minus the risk free rate
 - *smb*: return of a portfolio being long in small stocks and short in big stocks
 - *hml*: return of a portfolio being long in stocks with a high book-to-market ratio and short in stocks with a low book-to-market ratio
 - *rf*: risk free rate
- Stock price data for Tesla, *tesla.txt*
 - *permno*: number identifying Tesla
 - *date*: date
 - *PRC*: price
 - *RET*: return

You will assess whether Elon Musk's tweet significantly impacted Tesla's stock price using the standard event study methodology. Your analysis should proceed in following steps.

1. Define timing: Set the event date t^* . We will look at an event window that starts one day before, $k_b = 1$, and ends a day after, $k_a = 1$, the event day t^* . In order to avoid confounding events affecting our study, we will assume a hold-out window of 5 days, $h = 5$. We assume an estimation period of 250 days, $e = 250$. Please refer to the timeline below for clarification. In this context days means trading days. Note: It might be helpful if you declare a variable indicating the event, hold-out, and estimation window.



2. [Optional] Plot Tesla's stock price and mark the event date.
3. Compute abnormal return AR_t during the event window. Abnormal return is defined as the difference between observed return R_t and expected (or normal) return $E(R_t|X_t)$.

$$AR_t = R_t - E(R_t|X_t)$$

In order to estimate the latter, we use insights from finance theory. In particular, we use following return generating process (RGP):

- Fama-French 3-factor risk adjustment model: $(R_t - R_t^f) = \alpha + \beta_{mkt}(R_t^{mkt} - R_t^f) + \beta_{smb}SMB_t + \beta_{hml}HML_t + u_t$

Note that the Fama-French model uses excess returns $R_t - R_t^f$. Estimate the parameters of the RGP using only data of the estimation period and compute abnormal return during the event window.

4. Compute cumulative abnormal return CAR by summing abnormal return AR_t during the event window.
5. Test whether the abnormal event return is significantly different from 0. You can compute the t-value for following hypothesis test: $H_0: CAR = 0$ versus $H_1: CAR \neq 0$ as follows,

$$t.value = \frac{CAR}{\sqrt{l * \hat{\sigma}^2}}$$

l indicates the length of the event window and $\hat{\sigma}^2$ is the estimated variance of abnormal returns, using data from the estimation window. You can use $var(u_t)$ (this is the variance of the residual of the RGP model) as $\hat{\sigma}^2$.

6. Obviously the decision of the event day as well as the estimation window matter. Assume that the event window is only the day of the tweet, does this change the result?
7. [Optional] Yet another slightly simpler approach would be to estimate following time series regression model.

$$(R_t - R_t^f) = \alpha + \beta_1 * I(t = t^* - 1) + \beta_2 * I(t = t^*) + \beta_3 * I(t = t^* + 1) + \beta_{mkt}(R_t^{mkt} - R_t^f) + \beta_{smb}SMB_t + \beta_{hml}HML_t + u_t$$

using Tesla's return during the estimation, hold-out, and event window. I is a dummy variable turning one on specific days. Estimate this model and explain the intuition of why the model helps us to estimate returns associated with a given event.

9. What is your assessment: Did Elon Musk increase the stock price by tweeting his ambition to take the company private?

Note: the data is provided in two different dataframes. You can use the following link [<https://dplyr.tidyverse.org/reference/mutate-joins.html>] for some information on how to merger dataframes.

Note: you can work with dates as numerals to define the different time periods.

```

#-----#
#           SOLUTION           #
#-----#

# packages needed
require(dplyr)

# set the working directory
setwd("C:/Users/s13163/Dropbox/FIE401/data/data_labs")

# load the factor data
factor <- read.csv("factor_returns.txt")

# load Tesla's return data
tesla <- read.csv("tesla.txt", sep = "\t", stringsAsFactors = F)

# convenient adjustment
tesla <- tesla[order(tesla$date),]

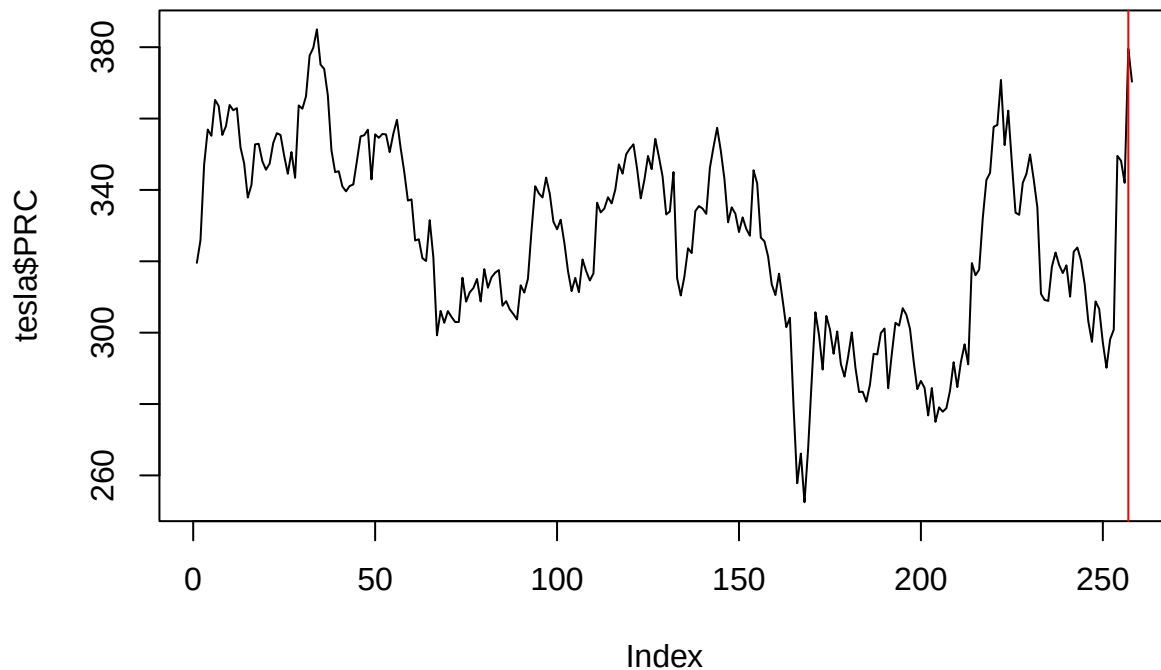
# merge
tesla <- left_join(tesla, factor)
rm(factor)

# define event periods
tesla$period <- NA
tesla$period[which(tesla$date == 20180807)+ -1:1] <- "event.period"
tesla$period[which(tesla$date == 20180807)+ -2:-6] <- "hold.out"
tesla$period[which(tesla$date == 20180807)+ -7:-256] <- "estimation.period"

# limit the data
tesla <- tesla[!is.na(tesla$period),]

# plot
plot(tesla$PRC, type = "l")
abline(v = which(tesla$date == 20180807), col="red")

```



```
# compute abnormal return
ff3_fit <- lm( (RET - rf) ~ mkt.rf + smb + hml,
              data = tesla[tesla$period == "estimation.period",])
tesla$ar_ff3 <- (tesla$RET-tesla$rf) - predict(ff3_fit, tesla)

# results -1 to 1 window
car_ff3 <- sum(tesla$ar_ff3[tesla$period == "event.period"])

var_ff3 <- var(tesla$ar_ff3[tesla$period == "estimation.period"])

tval_ff3 <- car_ff3 / sqrt(3 * var_ff3)
tval_ff3
```

```
## [1] 1.472728
```

```
# results on event day window
car_ff3 <- sum(tesla$ar_ff3[tesla$date == 20180807])

var_ff3 <- var(tesla$ar_ff3[tesla$period == "estimation.period"])

tval_ff3 <- car_ff3 / sqrt(1 * var_ff3)
tval_ff3
```

```
## [1] 4.576692
```

```
# Time series model
summary(lm(RET~rf ~ (tesla$date == 20180806) + (tesla$date == 20180807) +
           (tesla$date == 20180808)+ mkt.rf + smb + hml, data = tesla))
```

```
##
## Call:
## lm(formula = RET - rf ~ (tesla$date == 20180806) + (tesla$date ==
##      20180807) + (tesla$date == 20180808) + mkt.rf + smb + hml,
##      data = tesla)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.065665 -0.015032 -0.000459  0.011139  0.144408
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      -0.0006591   0.0015618   -0.422   0.67339
## tesla$date == 20180806TRUE -0.0277880   0.0248951   -1.116   0.26540
## tesla$date == 20180807TRUE  0.1048051   0.0248608    4.216 3.48e-05 ***
## tesla$date == 20180808TRUE -0.0206802   0.0248723   -0.831   0.40651
## mkt.rf             1.5018669   0.1994680    7.529 9.17e-13 ***
## smb                0.1967339   0.3276502    0.600   0.54876
## hml               -0.9960292   0.3512187   -2.836   0.00494 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.02481 on 251 degrees of freedom
## Multiple R-squared:  0.2693, Adjusted R-squared:  0.2518
## F-statistic: 15.42 on 6 and 251 DF, p-value: 4.835e-15
```